DBMS Mini Project

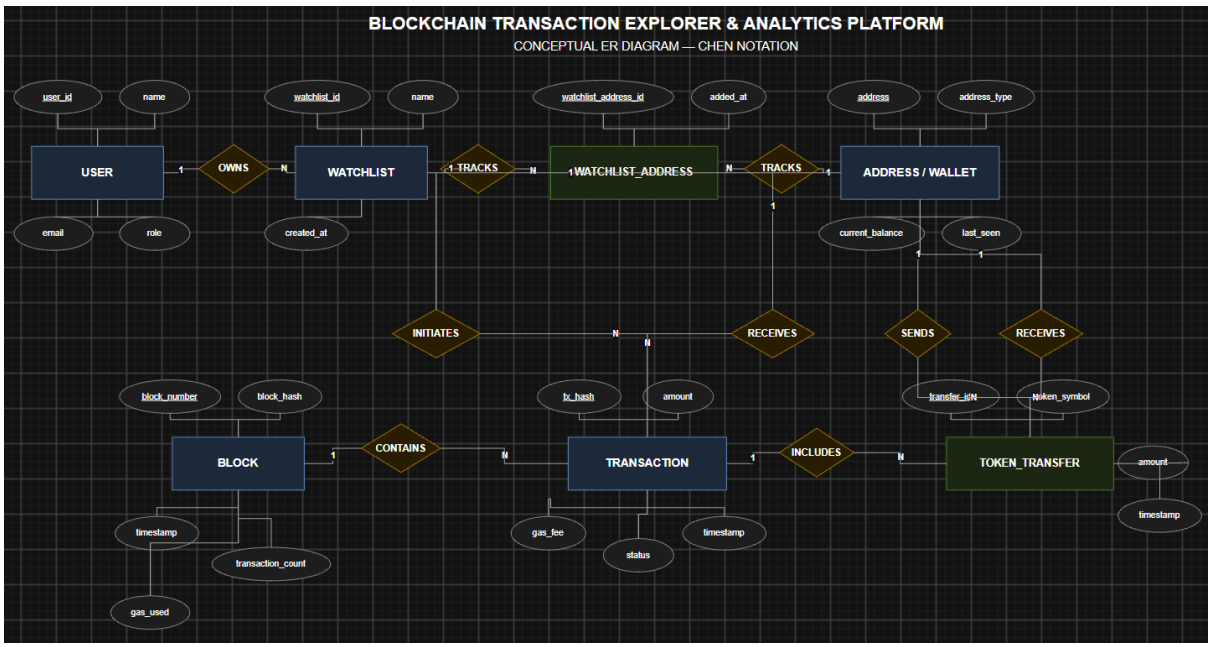
Abhiram Kolalapudi - PES2UG24CS916

Pavan Kumar K M - PES2UG24CS646

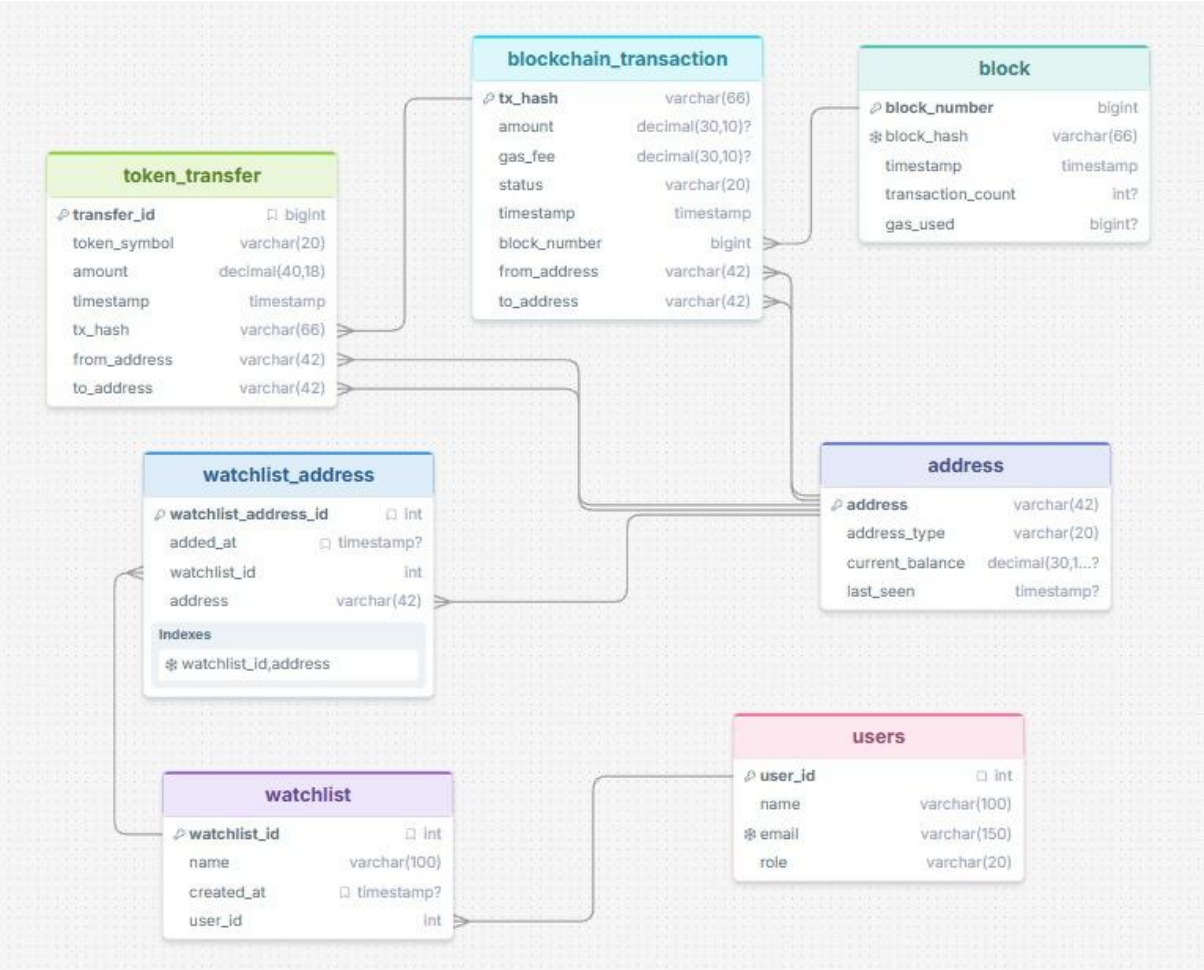
Sem-5 Section-K

Project Title: Blockchain Transaction Explorer & Analytics Platform

ER Diagram:



Schema Diagram:



DDL:

```
mysql> CREATE DATABASE blockchain_explorer;
Query OK, 1 row affected (0.03 sec)

mysql> use blockchain_explorer;
Database changed
mysql> CREATE TABLE users (
    ->     user_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     name VARCHAR(100) NOT NULL,
    ->     email VARCHAR(150) NOT NULL UNIQUE,
    ->     role VARCHAR(20) NOT NULL
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> desc users;
+---------+--------------+------+-----+---------+----------------+
| Field   | Type         | Null | Key | Default | Extra          |
+---------+--------------+------+-----+---------+----------------+
| user_id | int          | NO   | PRI | NULL    | auto_increment |
| name    | varchar(100) | NO   |     | NULL    |                |
| email   | varchar(150) | NO   | UNI | NULL    |                |
| role    | varchar(20)  | NO   |     | NULL    |                |
+---------+--------------+------+-----+---------+----------------+
4 rows in set (0.02 sec)
```

```
mysql> CREATE TABLE watchlist (
    ->     watchlist_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     name VARCHAR(100) NOT NULL,
    ->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     user_id INT NOT NULL,
    ->
    ->     FOREIGN KEY (user_id)
    ->         REFERENCES users(user_id)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> desc watchlist;
+-------------+-------------+------+-----+-------------------+-------------------+
| Field       | Type        | Null | Key | Default           | Extra             |
+-------------+-------------+------+-----+-------------------+-------------------+
| watchlist_id | int         | NO   | PRI | NULL              | auto_increment    |
| name        | varchar(100) | NO   |     | NULL              |                   |
| created_at  | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| user_id     | int         | NO   | MUL | NULL              |                   |
+-------------+-------------+------+-----+-------------------+-------------------+
4 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE address (
    ->     address VARCHAR(42) PRIMARY KEY,
    ->     address_type VARCHAR(20) NOT NULL,
    ->     current_balance DECIMAL(30,10) DEFAULT 0,
    ->     last_seen TIMESTAMP NULL
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> desc address;
+-----------------+----------------+------+-----+--------------+-------+
| Field           | Type           | Null | Key | Default      | Extra |
+-----------------+----------------+------+-----+--------------+-------+
| address         | varchar(42)    | NO   | PRI | NULL         |       |
| address_type    | varchar(20)    | NO   |     | NULL         |       |
| current_balance | decimal(30,10) | YES  |     | 0.0000000000 |       |
| last_seen       | timestamp      | YES  |     | NULL         |       |
+-----------------+----------------+------+-----+--------------+-------+
4 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE watchlist_address (
    ->     watchlist_address_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     added_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     watchlist_id INT NOT NULL,
    ->     address VARCHAR(42) NOT NULL,
    ->
    ->     FOREIGN KEY (watchlist_id)
    ->         REFERENCES watchlist(watchlist_id),
    ->
    ->     FOREIGN KEY (address)
    ->         REFERENCES address(address),
    ->
    ->     UNIQUE (watchlist_id, address)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> desc watchlist_address;
+---------------------+-------------+------+-----+-------------------+-------------------+
| Field               | Type        | Null | Key | Default           | Extra             |
+---------------------+-------------+------+-----+-------------------+-------------------+
| watchlist_address_id | int         | NO   | PRI | NULL              | auto_increment    |
| added_at            | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| watchlist_id        | int         | NO   | MUL | NULL              |                   |
| address             | varchar(42) | NO   | MUL | NULL              |                   |
+---------------------+-------------+------+-----+-------------------+-------------------+
4 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE block (
    ->     block_number BIGINT PRIMARY KEY,
    ->     block_hash VARCHAR(66) NOT NULL UNIQUE,
    ->     timestamp TIMESTAMP NOT NULL,
    ->     transaction_count INT DEFAULT 0,
    ->     gas_used BIGINT DEFAULT 0
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> desc block;
+-------------------+-------------+------+-----+---------+-------+
| Field             | Type        | Null | Key | Default | Extra |
+-------------------+-------------+------+-----+---------+-------+
| block_number      | bigint      | NO   | PRI | NULL    |       |
| block_hash        | varchar(66) | NO   | UNI | NULL    |       |
| timestamp         | timestamp   | NO   |     | NULL    |       |
| transaction_count | int         | YES  |     | 0       |       |
| gas_used          | bigint      | YES  |     | 0       |       |
+-------------------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE blockchain_transaction (
    ->     tx_hash VARCHAR(66) PRIMARY KEY,
    ->     amount DECIMAL(30,10) DEFAULT 0,
    ->     gas_fee DECIMAL(30,10) DEFAULT 0,
    ->     status VARCHAR(20) NOT NULL,
    ->     timestamp TIMESTAMP NOT NULL,
    ->
    ->     block_number BIGINT NOT NULL,
    ->     from_address VARCHAR(42) NOT NULL,
    ->     to_address VARCHAR(42) NOT NULL,
    ->
    ->     FOREIGN KEY (block_number)
    ->         REFERENCES block(block_number),
    ->
    ->     FOREIGN KEY (from_address)
    ->         REFERENCES address(address),
    ->
    ->     FOREIGN KEY (to_address)
    ->         REFERENCES address(address)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> desc blockchain_transaction;
+--------------+---------------+------+-----+-------------+-------+
| Field        | Type          | Null | Key | Default     | Extra |
+--------------+---------------+------+-----+-------------+-------+
| tx_hash      | varchar(66)   | NO   | PRI | NULL        |       |
| amount       | decimal(30,10)| YES  |     | 0.0000000000|       |
| gas_fee      | decimal(30,10)| YES  |     | 0.0000000000|       |
| status       | varchar(20)   | NO   |     | NULL        |       |
| timestamp    | timestamp     | NO   |     | NULL        |       |
| block_number | bigint        | NO   | MUL | NULL        |       |
| from_address | varchar(42)   | NO   | MUL | NULL        |       |
| to_address   | varchar(42)   | NO   | MUL | NULL        |       |
+--------------+---------------+------+-----+-------------+-------+
8 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE token_transfer (
    ->     transfer_id BIGINT PRIMARY KEY AUTO_INCREMENT,
    ->     token_symbol VARCHAR(20) NOT NULL,
    ->     amount DECIMAL(40,18) NOT NULL,
    ->     timestamp TIMESTAMP NOT NULL,
    ->
    ->     tx_hash VARCHAR(66) NOT NULL,
    ->     from_address VARCHAR(42) NOT NULL,
    ->     to_address VARCHAR(42) NOT NULL,
    ->
    ->     FOREIGN KEY (tx_hash)
    ->         REFERENCES blockchain_transaction(tx_hash),
    ->
    ->     FOREIGN KEY (from_address)
    ->         REFERENCES address(address),
    ->
    ->     FOREIGN KEY (to_address)
    ->         REFERENCES address(address)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> desc token_transfer;
+--------------+---------------+------+-----+---------+----------------+
| Field        | Type          | Null | Key | Default | Extra          |
+--------------+---------------+------+-----+---------+----------------+
| transfer_id  | bigint        | NO   | PRI | NULL    | auto_increment |
| token_symbol | varchar(20)   | NO   |     | NULL    |                |
| amount       | decimal(40,18)| NO   |     | NULL    |                |
| timestamp    | timestamp     | NO   |     | NULL    |                |
| tx_hash      | varchar(66)   | NO   | MUL | NULL    |                |
| from_address | varchar(42)   | NO   | MUL | NULL    |                |
| to_address   | varchar(42)   | NO   | MUL | NULL    |                |
+--------------+---------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```